LTX-Video: Realtime Video Latent Diffusion

Lightricks
Arxiv, 2024

딥러닝논문읽기모임 7기
이미지처리팀

안 종 식 최 비 결 최 승 준

CONTENTS

01

Introduction

01. Introduction

1. Video Generation

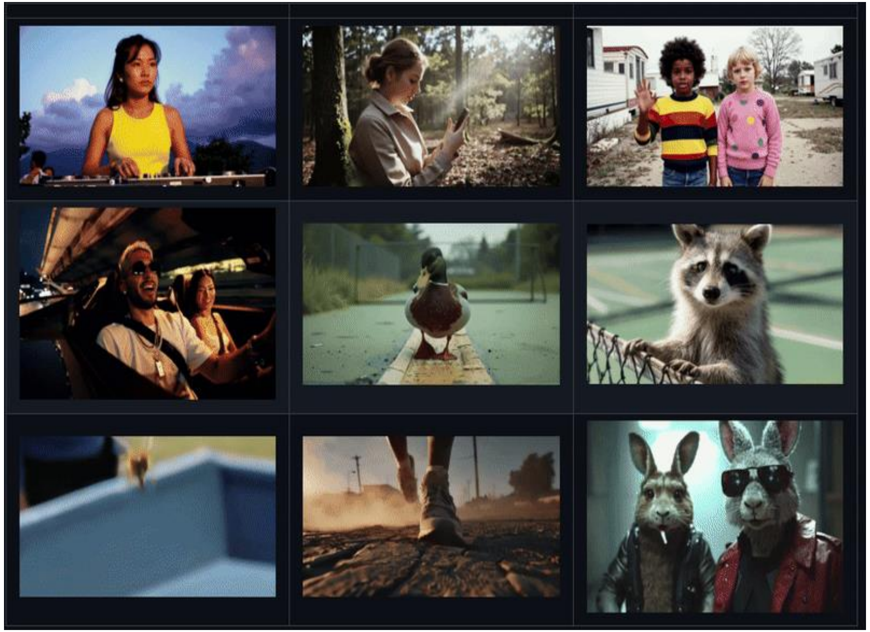
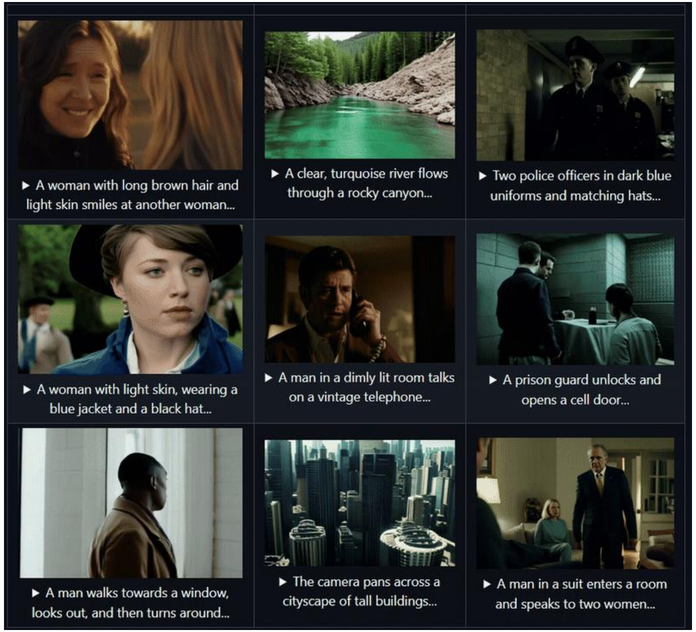


Image to Video Generation

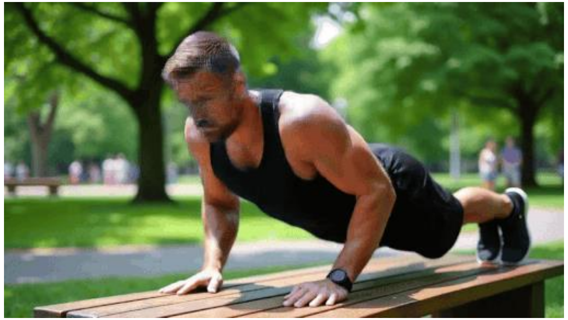


Text to Video Generation

출처 : https://github.com/Lightricks/LTX-Video

01. Introduction

1. Video Generation

Lightricks

768x512 Resolution, 25FPS Video, 5sec

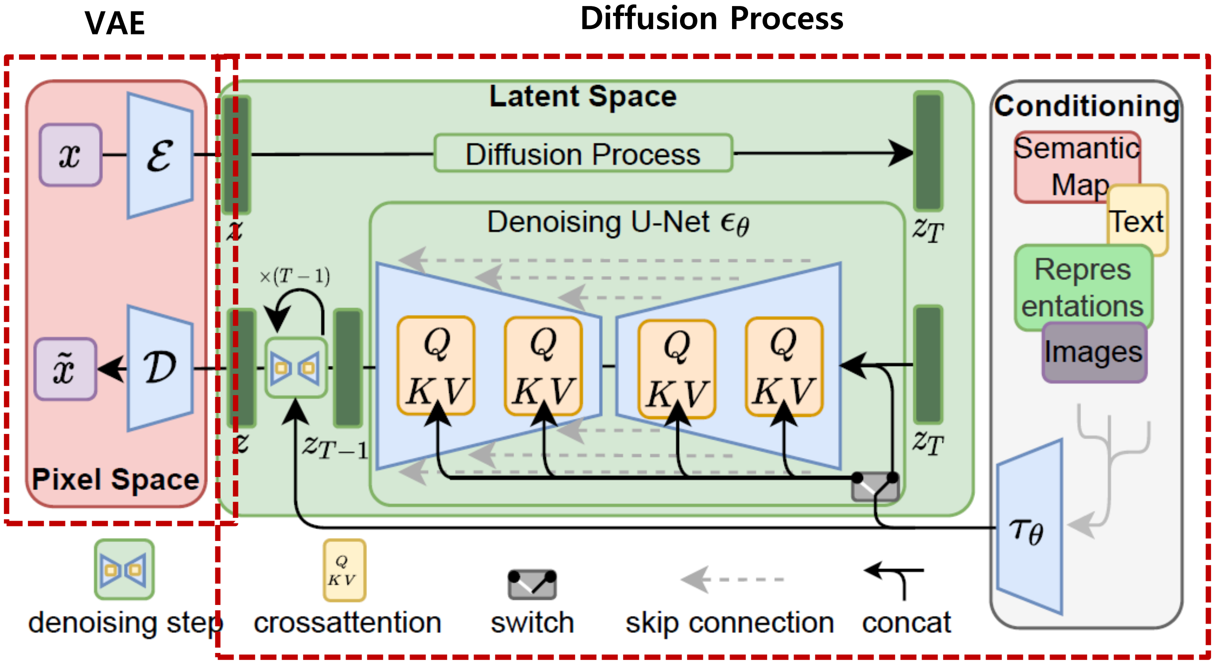
2sec

(GPU : H100, Diffusion Step : 20)

출처 : https://github.com/THUDM/CogVideo
출처 : https://github.com/hpcaitech/Open-Sora
출처 : Flaticon.com

01. Introduction

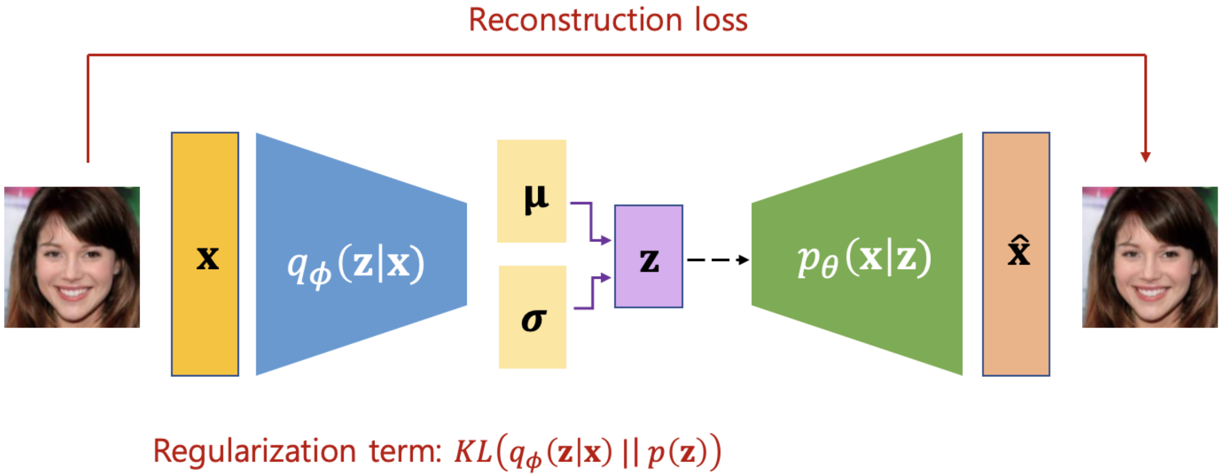
2. Diffusion Model : Stable Diffusion



출처 : High-Resolution Image Synthesis with Latent Diffusion Models

01. Introduction

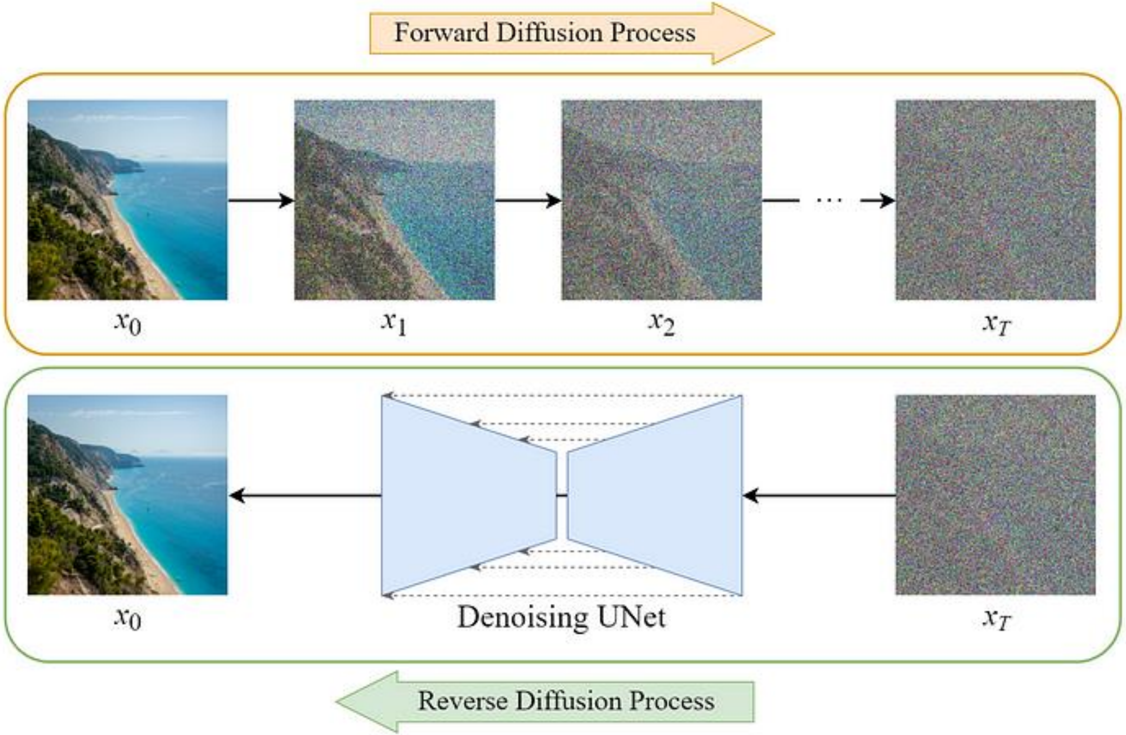
2. Diffusion Model : VAE



출처 : https://pasus.tistory.com/264

01. Introduction

2. Diffusion Model : Diffusion Process



출처 : https://xoft.tistory.com/32#google_vignette

02

Method

02. Method

1. Video VAE (3D VAE)

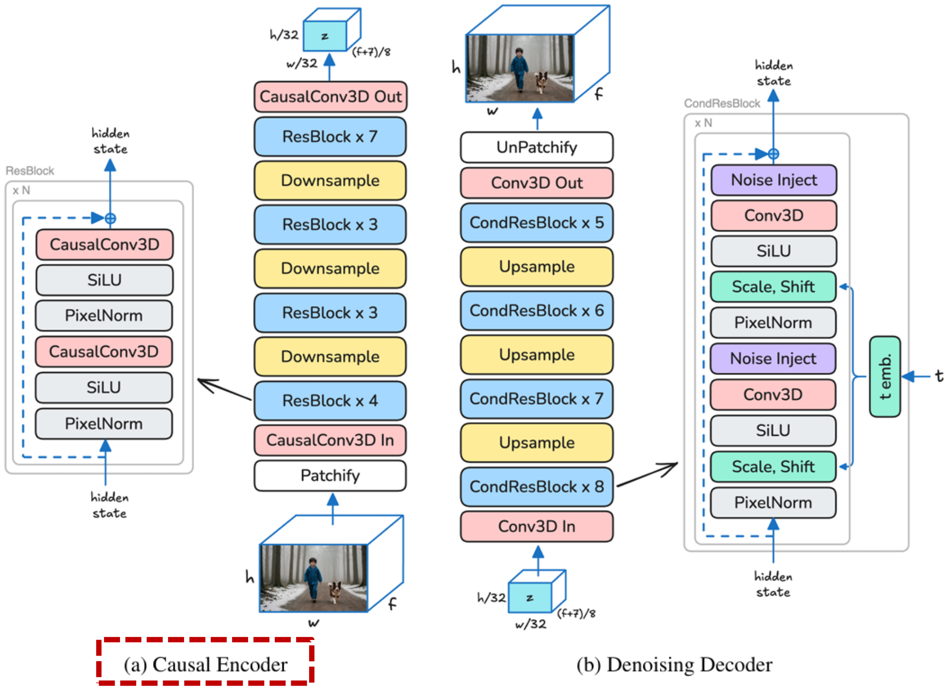


Figure 4: The *LTX-Video* Video-VAE architecture: (a) *Causal Encoder* utilizing 3D Causal Convolutions, applying $32 \times 32 \times 8$ compression (except the first frame, which is encoded as a separate latent frame). (b) *Denoising Decoder* with diffusion-timestep conditioning and multi-layer noise injection.

Casual Encoder

- 과거+현재 정보 활용하여 Encoding

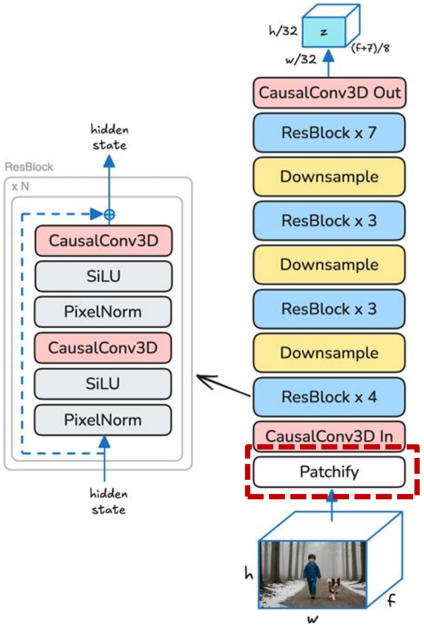
Non-casual Encoder

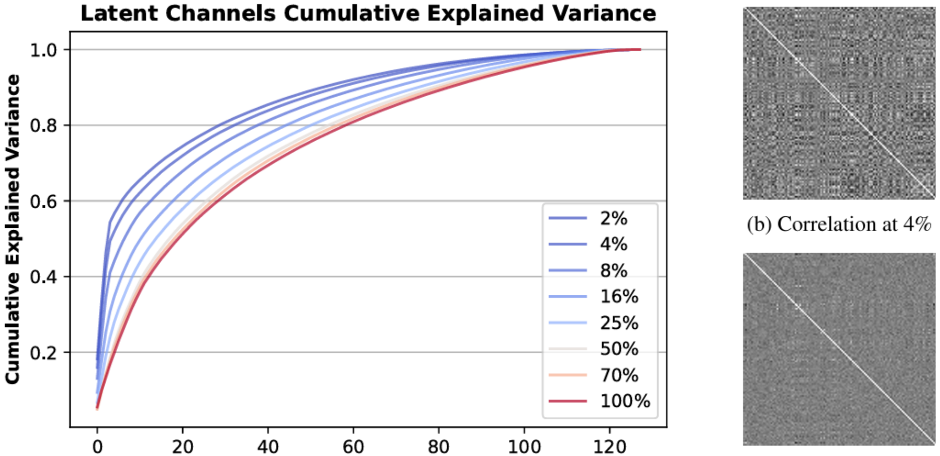
- 과거+현재+미래 정보 활용하여 Encoding

02. Method

1. Video VAE (3D VAE) : Encoder

Reducing Redundancy





(b) Correlation at 4%

(c) Final Correlation

(a) Latent channels cumulative explained variance at different training steps.

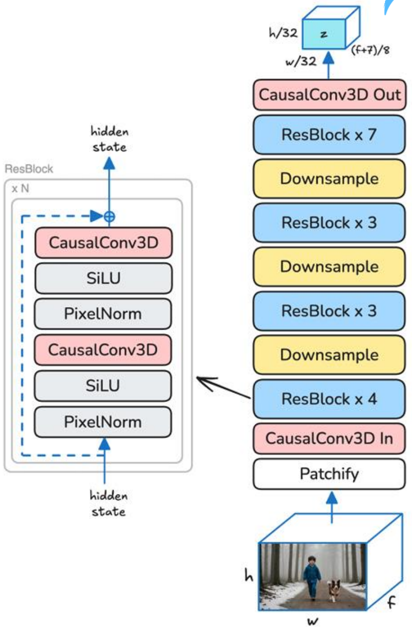
Figure 3: Latent-space redundancy. (a) Cumulative explained-variance of latent channels at different training steps (2% - 100% of training). As training progresses, the redundancy reduces and components contribute more evenly to the variance. (b, c) Latent channels auto-correlation matrices: high off-diagonal values early (at 4% of total training steps) and near-zero at training completion.

02. Method

1. Video VAE (3D VAE) : Encoder

High Compression Ratio

- **Spatial : 32 x 32 x 3 Compression**
- **Temporal : 8 Compression**
- **Spatiotemporal (Total) : 32 x 32 x 3 x 8 / 128(Latent Vector's Channel) = 192**



	MovieGen	HunyuanVideo	PyramidFlow	CogVideoX	LTX-Video
Base model size	30B	13B	2B	2B / 5B	1.9B
Transformer hidden dimension	6144	3072	1536	1920 / 3072	2048
FFN dimension factor	$2/3 \times 4$	4	4	4	4
VAE spatio-temporal compression	$8 \times 8 \times 8$	$8 \times 8 \times 4$	$8 \times 8 \times 8$	$8 \times 8 \times 4$	$32 \times 32 \times 8$
VAE output channels	16	16	16	16	128
VAE total compression	1:96	1:48	1:96	1:48	1:192
Transformer input patchifier	$2 \times 2 \times 1$	$2 \times 2 \times 1$	$2 \times 2 \times 1$	$2 \times 2 \times 1$	$1 \times 1 \times 1$
Tokens to pixels ratio	1:2048	1:1024	1:2048	1:1024	1:8192
Number of Transformer blocks	48	60	24	30 / 42	28
Attention Blocks Architecture	Self + Cross	Self-only	Self-only	Self-only	Self + Cross
Upsampling model	7B params	N/A	N/A	N/A	N/A

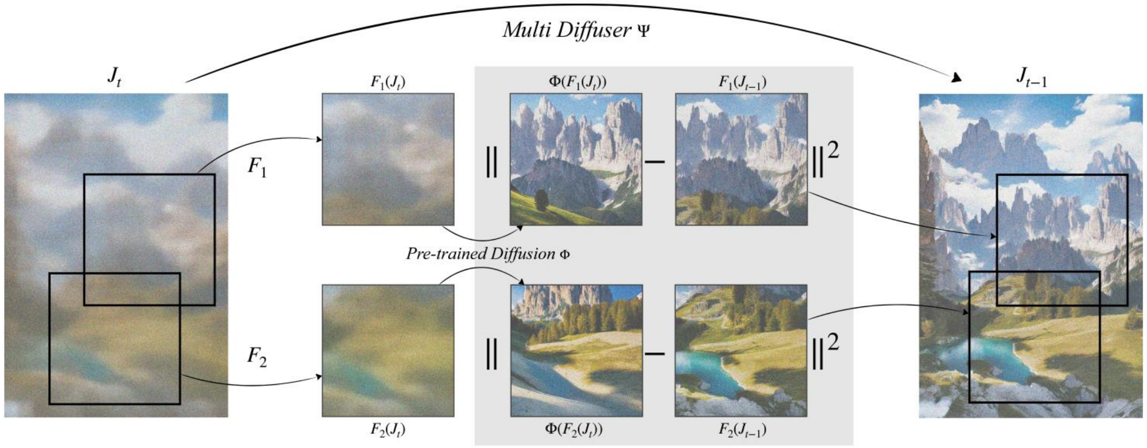
Table 1: Comparison of Model Specifications.

02. Method

1. Video VAE (3D VAE) : Decoder

Enhancing high-frequency details Second Stage Diffusion

Improved temporal consistency with Multi-Diffusion. Memory constraints prohibit us from training the Spatial Upsampler on longer video durations. As a result, during inference, we upsample videos in a sliding window fashion resulting in noticeable inconsistencies at the boundaries. To prevent this, we leverage MultiDiffusion (Bar-Tal et al., 2023), a training-free optimization that ensures consistency across different generation processes bound by a common set of constraints. Specifically, we use a weighted average of the latents from overlapping frames in each denoising step, facilitating the exchange of information across consecutive windows to enhance temporal consistency in the output.



🧠 예시로 보면?

python 복사 편집

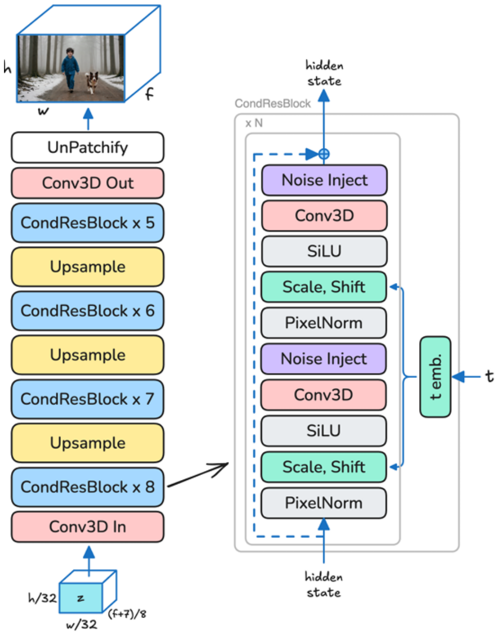
```python
z = torch.randn(latent_shape, requires_grad=True)
optimizer = torch.optim.Adam([z], lr=1e-2)

for t in timesteps:
    loss = 0
    for crop in crops:
        eps_pred = model(z_crop, t, cond)
        loss += ((eps_pred - eps_target)**2).mean()

    optimizer.zero_grad()
    loss.backward()  # ⛔ model 파라미터는 고정, z만 업데이트
    optimizer.step()
```

출처 : Movie Gen: A Cast of Media Foundation Models

02. Method

1. Video VAE (3D VAE) : Decoder

Enhancing high-frequency details



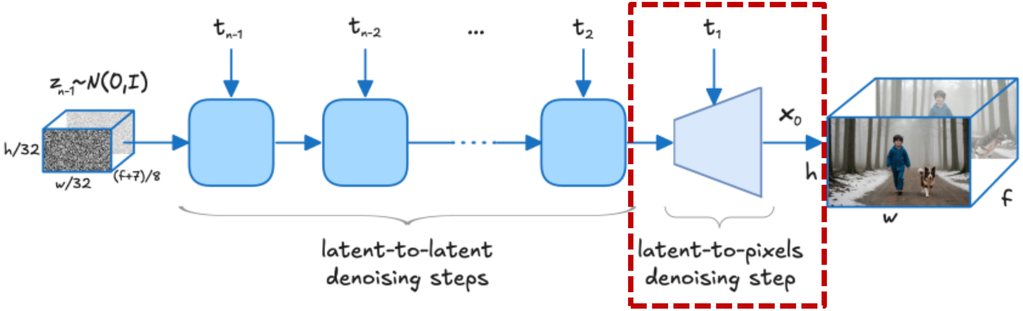
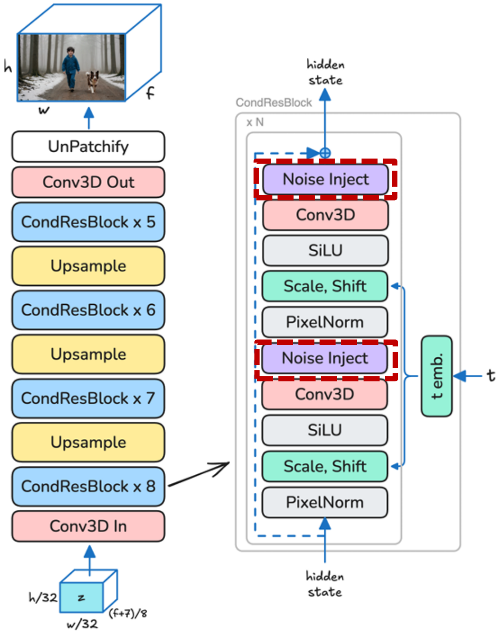


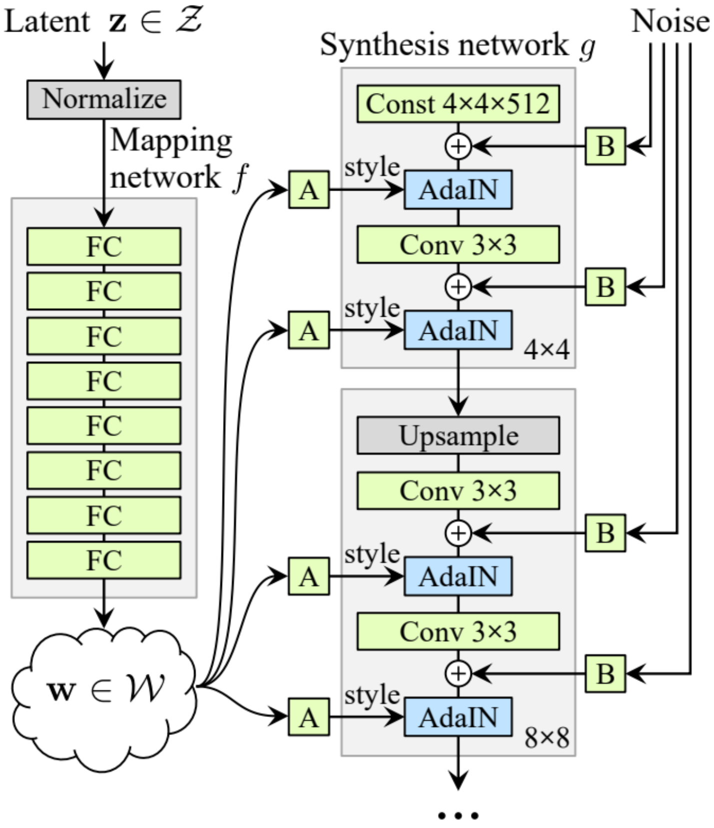
Figure 2: *LTX-Video* holistic denoising strategy – latent-to-latent diffusion denoising steps + final latent-to-pixels denoising step.

02. Method

1. Video VAE (3D VAE) : Decoder

Enhancing high-frequency details : Noise Injection



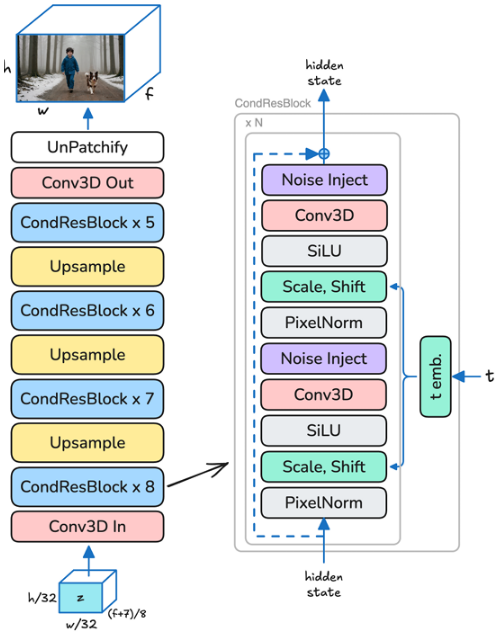


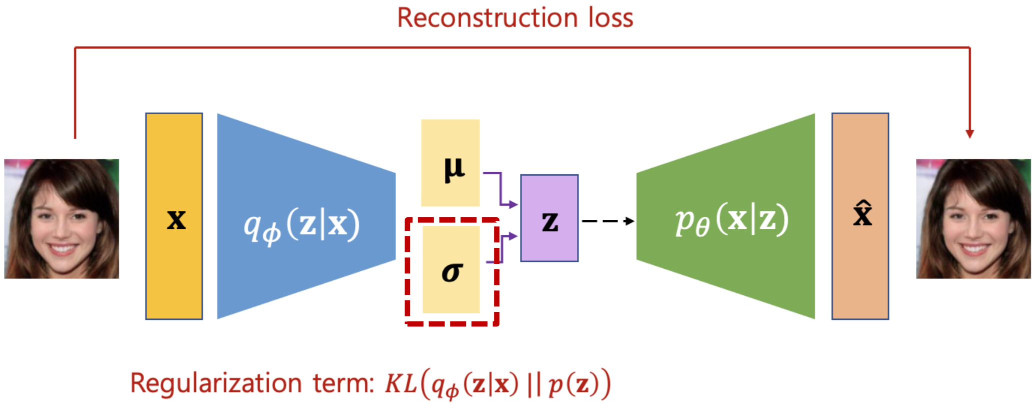
출처 : A Style-Based Generator Architecture for Generative Adversarial Networks

02. Method

1. Video VAE (3D VAE) : Decoder

Reducing Redundancy





출처 : https://pasus.tistory.com/264

02. Method

1. Video VAE (3D VAE) : Loss

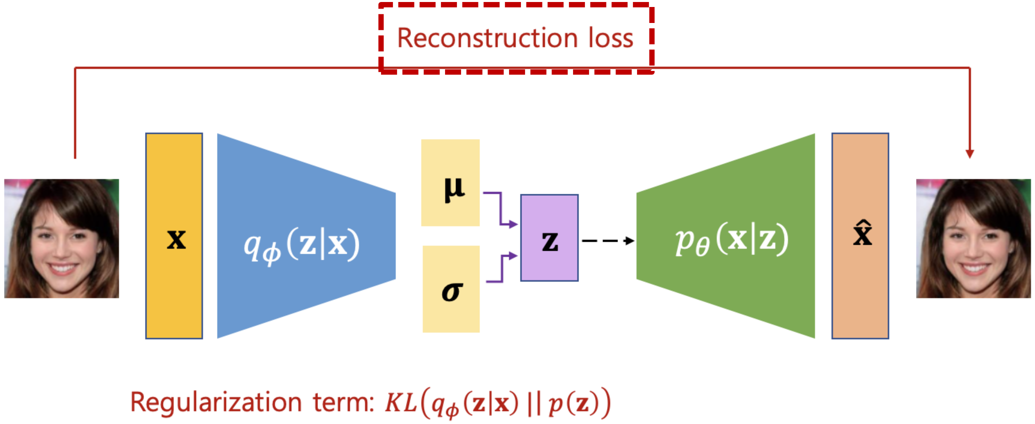
- Pixel Reconstruction Loss (MSE)
- Video DWT Loss
- Perceptual Loss (LPIPS)
- Reconstruction GAN Loss

출처 : Flaticon.com

02. Method

1. Video VAE (3D VAE) : Loss

Pixel Reconstruction Loss (MSE)

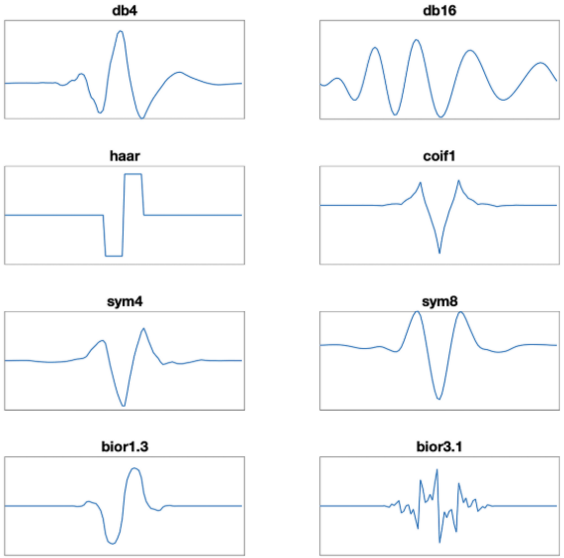


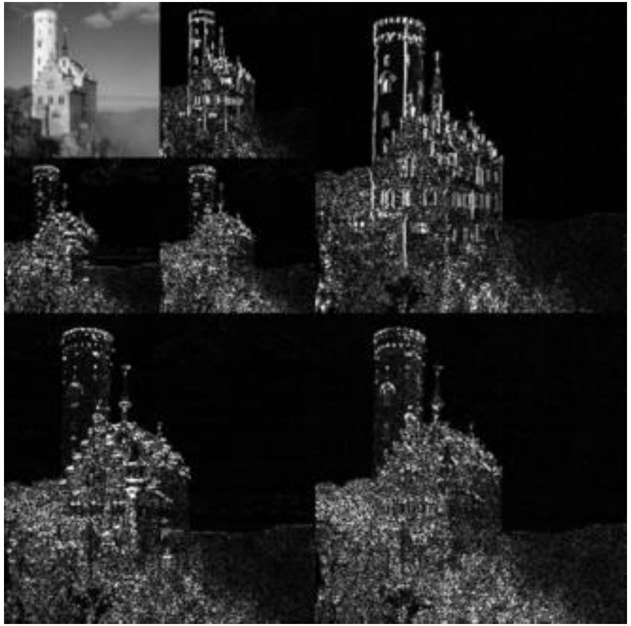
출처 : https://pasus.tistory.com/264

02. Method

1. Video VAE (3D VAE) : Loss

Video DWT Loss



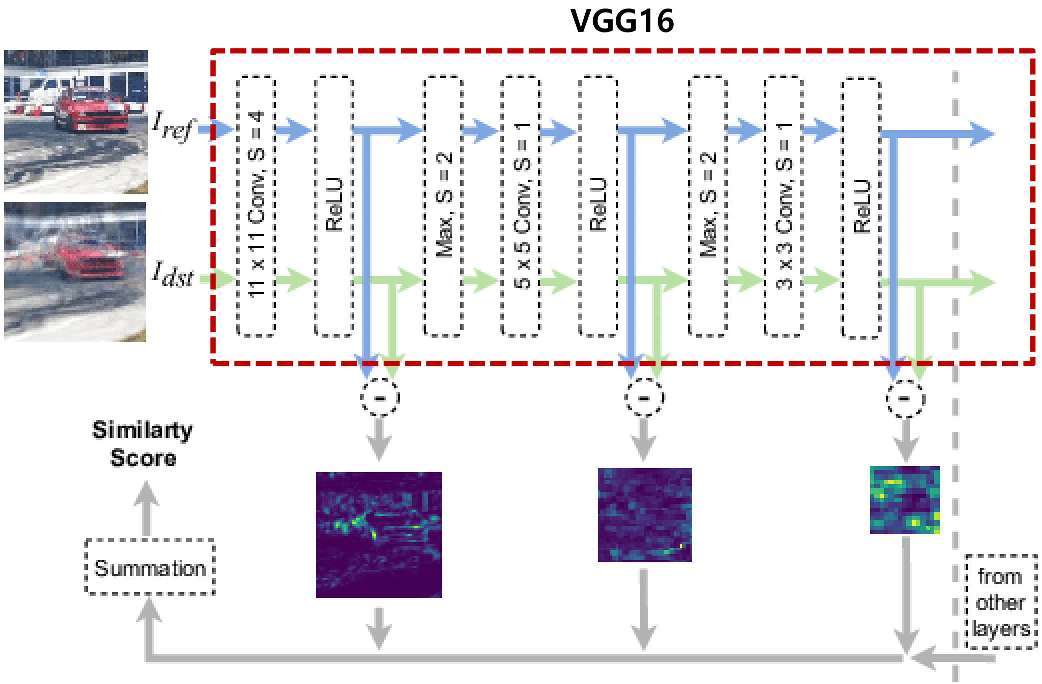


이미지1 출처 : https://ralasun.github.io/signal_analysis/2021/06/21/stft-dwt/
이미지2 출처 : https://en.wikipedia.org/wiki/Discrete_wavelet_transform

02. Method

1. Video VAE (3D VAE) : Loss

Perceptual Loss (LPIPS)

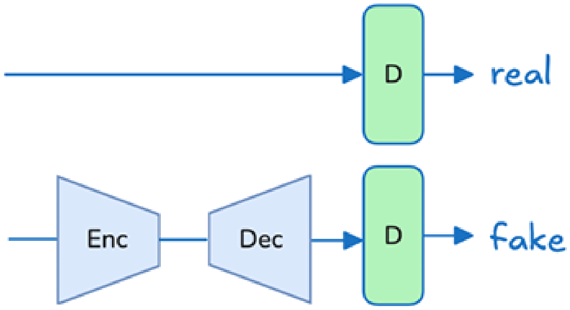


출처 : Shift-tolerant Perceptual Similarity Metric

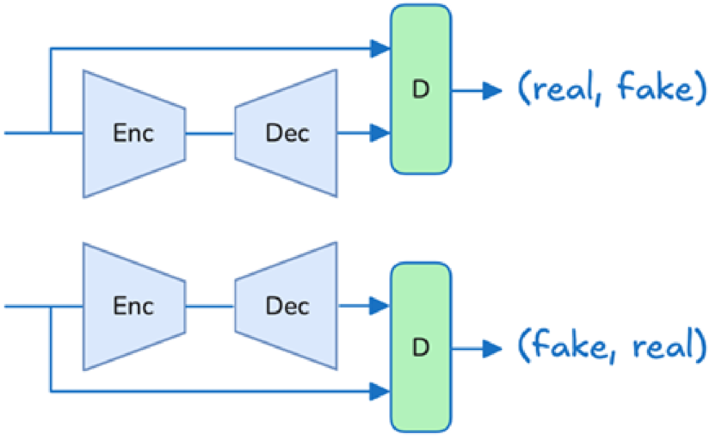
02. Method

1. Video VAE (3D VAE) : Loss

Reconstruction GAN Loss



(a) Traditional GAN

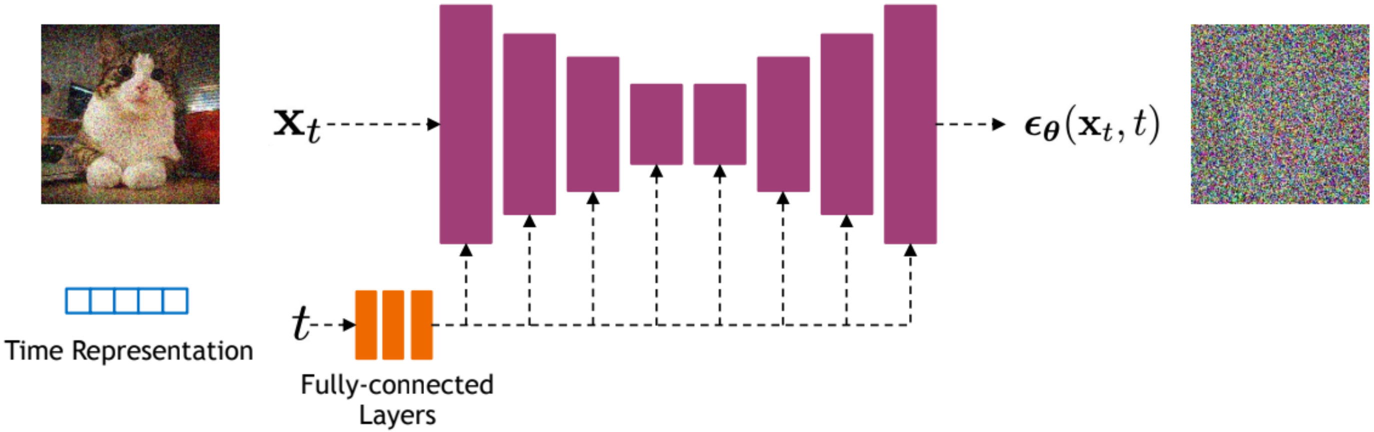


(b) Reconstruction GAN

02. Method

2. Video Transformer (Diffusion Process)

General Diffusion Model (U-Net)

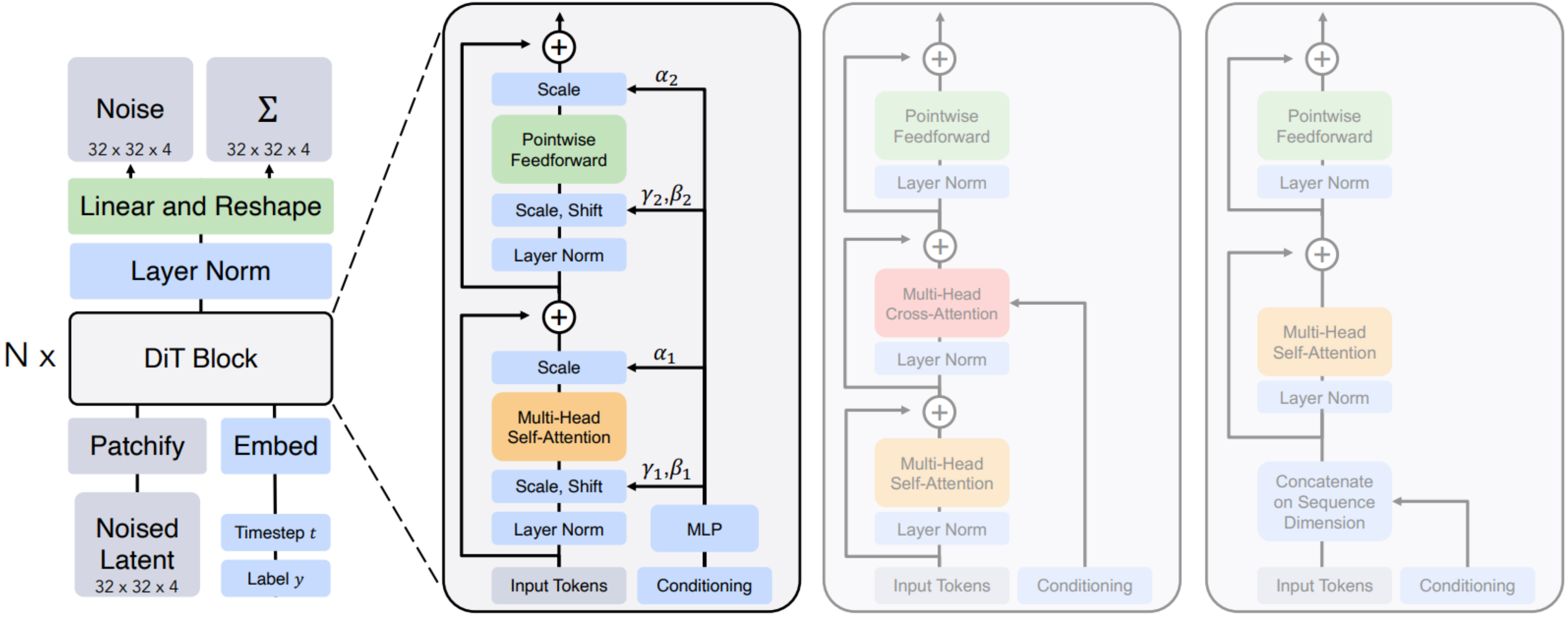


출처 : https://generativeai.pub/denoising-diffusion-probabilistic-models-from-scratch-728df8228565

02. Method

2. Video Transformer

DiT : Diffusion Transformer



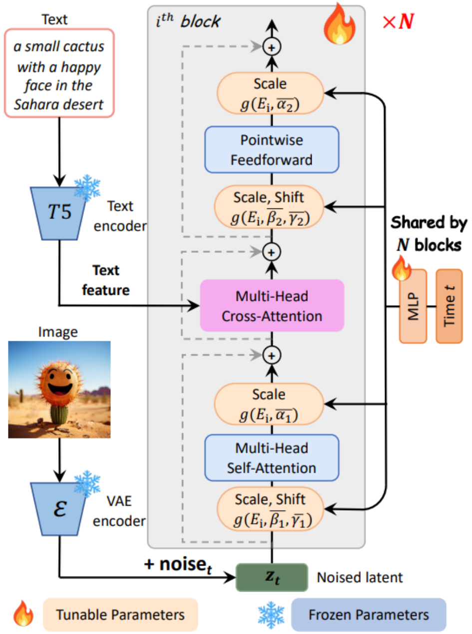
Latent Diffusion Transformer DiT Block with adaLN-Zero DiT Block with Cross-Attention DiT Block with In-Context Conditioning

출처 : Scalable Diffusion Models with Transformers

02. Method

2. Video Transformer

Pixart-α



LTX-Video

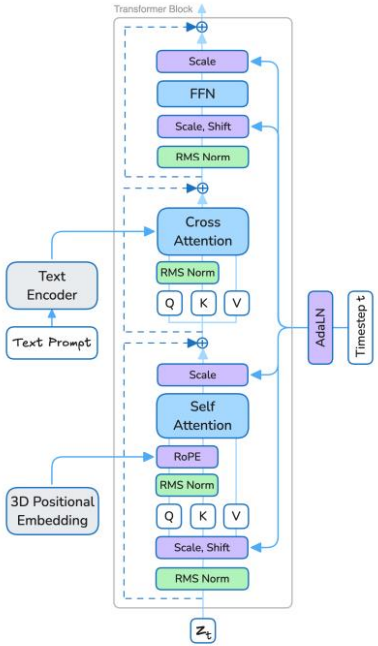


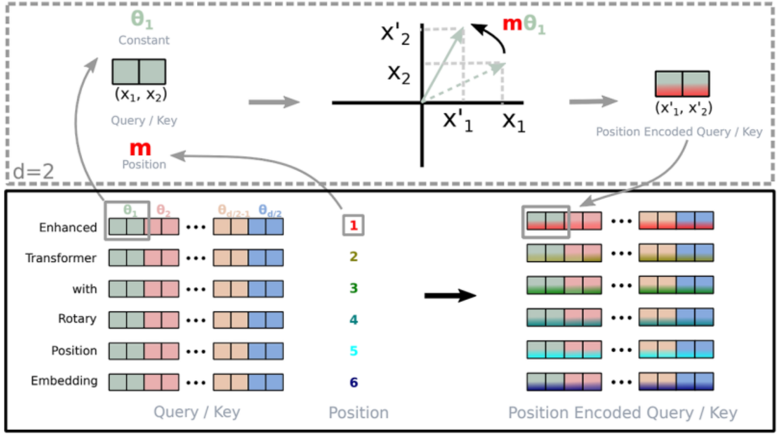
Figure 6: The *LTX-Video* 3D transformer-block architecture. Our architecture builds upon Pixart-α [8], replacing LayerNorm with RMSNorm and incorporating QK-normalization and RoPE positional embeddings.

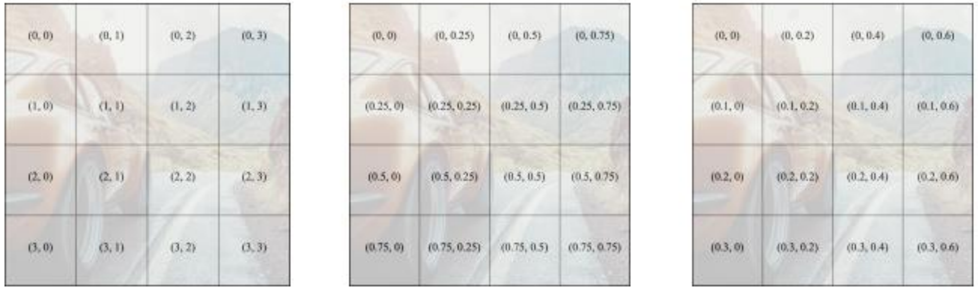
출처 : PixArt-α: Fast Training of Diffusion Transformer for Photorealistic Text-to-Image Synthesis

02. Method

2. Video Transformer

Rotational Positional Embedding (RoPE)





(a) (b) (c)

- **Absolute positional Embeddings**
- **RoPE with Fractional Coordinates**
- **RoPE with fractional coordinates normalized by predefined maximum coordinates**

출처 : https://ostin.tistory.com/463

02. Method

2. Video Transformer

Image Conditioning

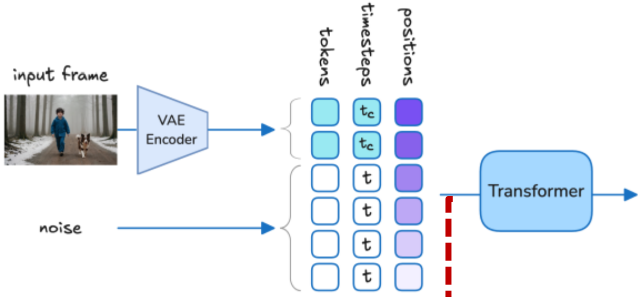


Figure 9: *LTX-Video* image-to-video inference pipeline – first-frame conditioning. The diffusio
timestep and corresponding noise level is defined per-token. For example, conditioning tokens ca
have diffusion timesteps of $t_c = 0$ and contain un-noised encoded tokens.

Text Conditioning & QK Normalization

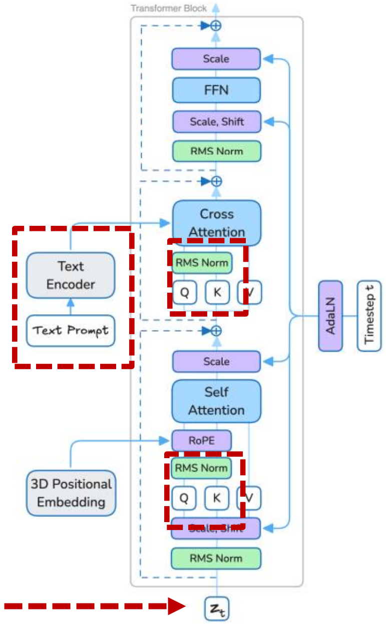


Figure 6: The *LTX-Video* 3D transformer-block architecture. Our architecture builds upon Pixart-α [8], replacing LayerNorm with RMSNorm and incorporating QK-normalization and RoPE positional embeddings.

02. Method

2. Video Transformer

Timestep Scheduling

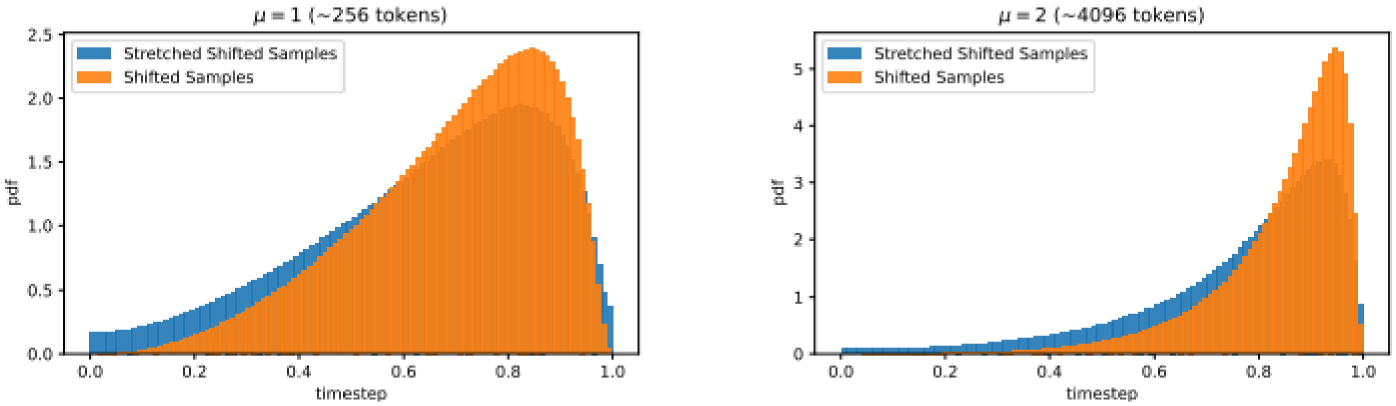


Figure 10: Timestep t sampling distribution during training, shown with two shifting parameters μ. We use the distributions shown in blue, which prevent near-zero probability at the tails.

03

Experiments

03. Experiments

1. Data Preparation

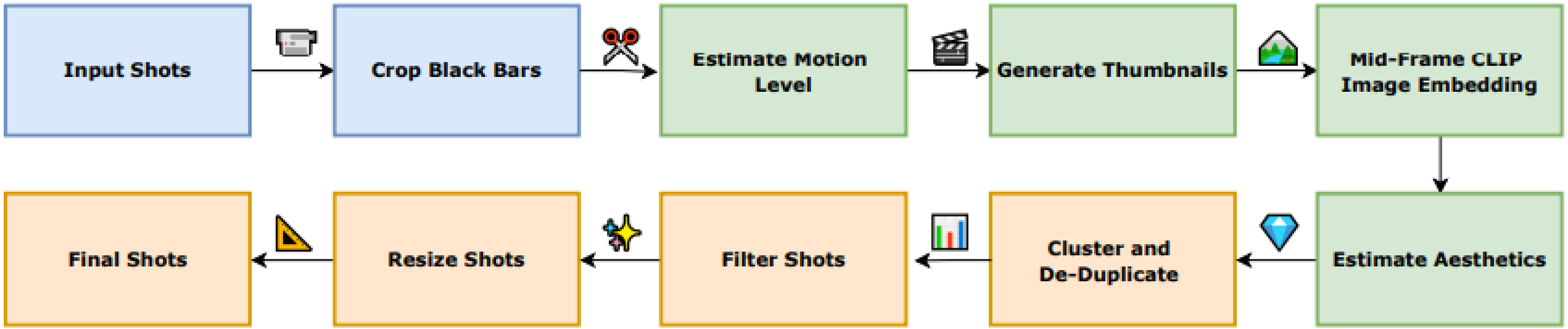


Figure 11: Our data processing pipeline.

03. Experiments

1. Data Preparation

Quality Control and Filtering

- Aesthetic Model 도입

Fine-tuning with Aesthetic Content

- Pretraining 후 Fine tuning 시 Filtering 한 영상으로만 학습

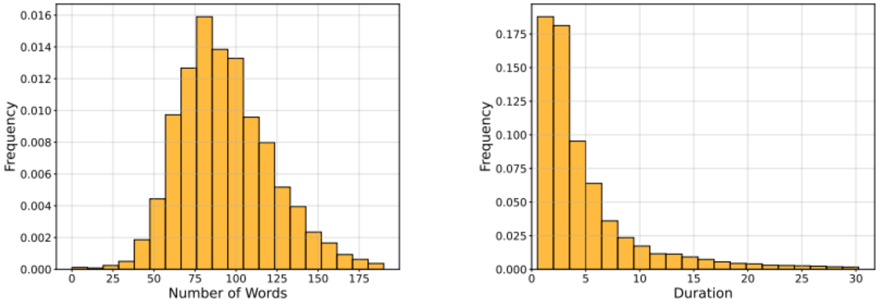
Motion and Aspect Ratio Filtering

- 움직임이 많은 동적인 이미지로 Filtering

Captioning and Metadata Enhancement

- Automatic image and Video Captioner

Figure 13: A word cloud of our captioned training data.



(a) Number of words per caption (b) Clip duration (seconds)

Figure 14: Training data statistics.

A large, colorful dragon puppet is carried down a street by a crowd of people. The dragon puppet is predominantly white with blue, green, and red accents, and it has large, expressive eyes and a long, serpentine body; it is suspended from a pole held by a person in the crowd, and its head moves slightly as it is moved down the street. The crowd is composed of people of all ages, many of whom are wearing winter clothes; they are cheering and waving their arms in the air, and some are taking pictures or videos with their phones. The street is lined with buildings, some of which have signs in Chinese characters; there are also a few trees and streetlights visible. The camera follows the dragon puppet from behind as it moves down the street, and it pans slightly to capture the surrounding crowd and the buildings; the lighting is natural and slightly overcast, casting soft shadows on the street. The scene is captured in real-life footage.

A Dalmatian dog stands on a window ledge, looks around, and then jumps. The dog has black spots on its white fur and is standing on a window ledge of a yellow building with green shutters. The dog looks to the left and then jumps from the window ledge. The camera is positioned across the street from the dog and remains stationary throughout the video. The lighting is natural sunlight, and the colors in the video are bright and vibrant. The scene is captured in real-life footage.

Figure 12: Our video captioner – input video (generated by Sora [1]) and our generated captions.

03. Experiments

2. Results

MovieGen 방식을 따름 (Human Survey Protocol, Win rate)

3.5 Evaluation

In this section, we explain how we evaluate the text-to-video quality of Movie Gen Video and other models. Our goal is to establish clear and effective evaluation metrics that identify a model's weaknesses and provide reliable feedback. We explain the different text-to-video evaluation axes and their design motivations in Section 3.5.1. We introduce our new benchmark, Movie Gen Video Bench, in Section 3.5.2. Throughout this work, we use human evaluation to assess the quality of generated videos across various evaluation axes. When evaluating each axis, we conduct pairwise A/B tests where expert human evaluators assess two videos side-by-side. Evaluators are instructed to choose a winner based on the axis being measured, or to declare a tie in case of no clear winner. We include a discussion on the motivation and reliability of using human evaluations and on existing automated metrics in Section 3.5.3.

- **참가자 : 총 20명**
- **평가 대상 : Open-sora Plan, CogVideoX, PyramidFlow, LIX-Video**
- **Metric : Win-rate**
- **기준**
  - ✓ 시각 품질
  - ✓ 모션 충실도
  - ✓ 프롬포트 일치도

- **Text-to-Video : Prompt 1,000개 사용**
- **Image-to-Video : FLUX.1로 생성한 이미지 + Prompt 1,000개**
- **동일 해상도, 길이, 생성 스탭 사용**
  - ✓ 768x512, 24FPS, 5초
  - ✓ Diffusion Step : 40

출처 : Movie Gen: A Cast of Media Foundation Models

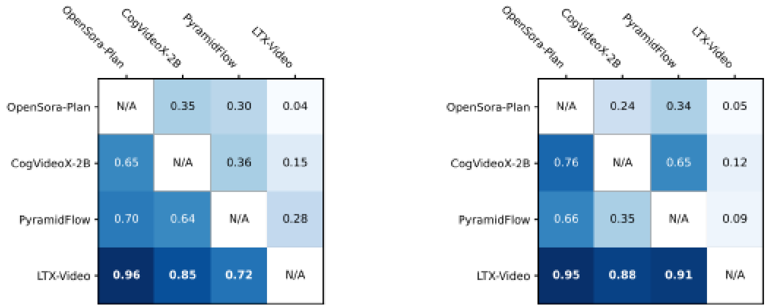
03. Experiments

2. Results

MovieGen 방식을 따름 (Human Survey Protocol, Win rate)

Table 2: Survey results showing the percentage of tests in which each model won for text-to-video and image-to-video tasks.

	Open-Sora Plan	CogVideoX 2B	PyramidFlow	*LTX-Video* (Ours)
Text-to-video	20%	38%	51%	**85%**
Image-to-video	20%	47%	35%	**91%**



(a) Text-to-Video (b) Image-to-Video

Figure 15: Pairwise performance matrix: each row indicates the win ratios of a model against all other models.

03. Experiments

3. Ablation

Reconstruction GAN vs. Traditional GAN

Figure 16: VAE reconstruction example – input frame (left), reconstruction with standard GAN loss (center), improved reconstruction with our proposed *Reconstruction GAN* loss (right)

03. Experiments

3. Ablation

Denoising VAE Decoder

4.4 Denoising VAE Decoder

Our holistic approach tasks the VAE decoder with performing the last denoising step in conjunction with converting the latents into pixels. To validate this design choice, we performed an internal user study comparing videos generated according to our approach to videos generated with the common approach, where denoising is performed solely by the diffusion-transformer, in latent space.

For the first set of results, our VAE decoder was conditioned on timestep $t = 0.05$. For the second set, the VAE decoder was conditioned on timestep $t = 0.0$ and did not perform any denoising.

The survey results indicated that videos generated by our method were strongly preferred over the standard results. The improvement was particularly evident in high-motion videos, where artifacts caused by the strong compression were mitigated by the VAE decoder's last-step denoising.

04

Conclusion

04. Conclusion

Contribution

- 고품질 영상 생성
- 매우 빠른 Diffusion Model
- 데이터셋 준비에 대한 준비성

Limitation

- Prompt 민감 (충실성으로 해석 가능)
- 장시간 비디오 지원 불가
- Background Knowledge 필수, 실험 내용 미공개

"A romantic couple is dancing gracefully at the beach during twilight. The man is wearing a black suit and white shirt, and the woman is in a flowing black dress, holding a bouquet of red flowers. The couple looks at each other lovingly, smiling warmly as gentle waves touch their feet. The scene is cinematic, softly lit with a dreamy blue hue. The camera slowly circles them as they dance barefoot in the sand, preserving their facial features in high detail."

04. Conclusion

Discussion

▶ 정량적인 지표로 신뢰성 있는 생성형 모델 평가 방법 ?

Pixel Level

$$\mathrm{MSE} = \frac{1}{N}\sum_{i=1}^{N}(x_i - \hat{x}_i)^2$$

$$\mathrm{PSNR} = 10\log_{10}\frac{\mathrm{MAX}^2}{\mathrm{MSE}}$$

분포기반

$$\mathrm{FID} = \|\mu_r - \mu_g\|^2 + \mathrm{Tr}(\Sigma_r + \Sigma_g - 2(\Sigma_r\Sigma_g)^{1/2})$$

✓ FVD (Frechet Video Distance)

✓ IS (Inception Score)

Perceptual

✓ CLIP Score / R-precision

Time Consistency

✓ tFID / S-VideoFID

$$\mathrm{TCM} = \frac{1}{T-1}\sum_{t}\|\mathrm{Flow}(x_t, x_{t+1}) - \mathrm{Flow}(\hat{x}_t, \hat{x}_{t+1})\|_1$$

Q & A

Thank you for your attention